Case Study of Evaluation of Advanced Public Transportation Systems by Multi-agent Simulation

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Abstract

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1 Introduction

This paper try to investigate the usability of transportation system improves when the number of demand buses, the on-demand transportation system, increases in Hakodate city, a local middle-sized city, in Japan with a traffic simulator.

On-demand type transportation systems have been attracting the attention as a solution to an issue of increase of transportation-handicapped people such as aged people who can not manipulate a vehicle. Such situation is quite serious in japan in recent years because of motorized and super-aging society.

The purpose of this work is to show the possibility to seamless transition from the traditional transportation system to advanced on-demand type one by operating both transportation in a mixed way. For this, we modeled two transportation systems, on-demand system (demand buses), and current system (route buses), then, we simulated a situation in which the two systems were mixed in Hakodate city.

2 Experimental Results

We modeled a full-demand bus, demand bus[1], as an on-demand transportation system. In the system, a demand of each passenger is assined to a bus by *successive best insertion method*[1]. We also modeled a route bus as a traditional transportation system. Twenty routes and schedules of actual public buses for weekdays in Hakodate are used for the operation of the route buses. The operation hours are assumed to be 5AM to 8PM. A demand occurs every 5 minutes and its origin and destination are distributed randomly in Hakodate area.

For the comparison of effects of two bus systems, we changed of the ratio of demand buses in a fixed number of all buses. The passengers choose the demand bus in the same ratio of the demand bus. The usability of the busses is defined as a ratio of the acutual travel time compared with a travel time in the case of on-foot.

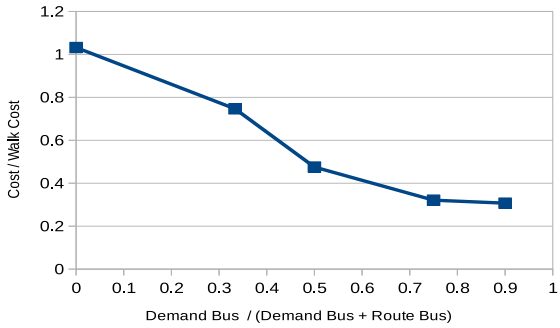


Figure1: The more the demand buses number increases, the more the trip time decreases.

The result is shown in Figure 1. This graph shows the changes of usability according to increase of the ratio of demand buses. This result tells the usability of total bus system will be improved by increasing the number of demand buses. This indicates the demand buses can be flexible enough in time scheduling and re-routing than route buses, so that the gradual changes by mixed operation will be possible.

References

[1] I. Noda, M. Ohta, K. Shinoda, Y. Kumadani, and H. Nakashima, "Evaluation of Usability of Dial-a-Ride Systems by Social Simulation", Multi-Agent-Based Simulation III. 4th International Workshop, MABS 2003 (LNAI-2927), pp.167-181 (2003).